# STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-SURJECTIVE-FLAT-LOCALLY-FINITE-PRESENTATION

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 3714-3728 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 3730-3773, the complete author proof (42 body lines).

*Editorial note.* Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 10 of queue_algebraic-stacks-sheaves.json (pb=2050, stmt 3714-3728, proof 3730-3773).

*Editorial note.* Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-surjective-flat-locally-finite-presentation' also exists in stacks-algebraic-stacks-morphisms — a different proposition in a different chapter file.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 055V: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3773–3773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03XQ: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3773–3773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03MN: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3773–3773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03ZD: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3773–3773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03MG: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3773–3773. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0467: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3773–3773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0468: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3773–3773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 04XX: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3773–3773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of categories fibred in groupoids over $(Sch/S)_{fppf}$.*

1. *Assume that $f$ is representable by algebraic spaces, surjective, flat, and locally of finite presentation. Then for any object $y$ of $\mathcal{Y}$ there exists an fppf covering $\{y_i \to y\}$ and objects $x_i$ of $\mathcal{X}$ such that $f(x_i) \cong y_i$ in $\mathcal{Y}$.*
2. *Assume that $f$ is representable by algebraic spaces, surjective, and smooth. Then for any object $y$ of $\mathcal{Y}$ there exists an étale covering $\{y_i \to y\}$ and objects $x_i$ of $\mathcal{X}$ such that $f(x_i) \cong y_i$ in $\mathcal{Y}$.*

*Proof.* Proof of (1). Suppose that $y$ lies over the scheme $V$. We may think of $y$ as a morphism $(Sch/V)_{fppf} \to \mathcal{Y}$. By definition the 2-fibre product $\mathcal{X} \times_{\mathcal{Y}} (Sch/V)_{fppf}$ is representable by an algebraic space $W$ and the morphism $W \to V$ is surjective, flat, and locally of finite presentation. Choose a scheme $U$ and a surjective étale morphism $U \to W$. Then $U \to V$ is also surjective, flat, and locally of finite presentation (see Morphisms of Spaces, Lemmas 0467, 0468, 03MG, 03XQ, and 03MN). Hence $\{U \to V\}$ is an fppf covering. Denote $x$ the object of $\mathcal{X}$ over $U$ corresponding to the 1-morphism $(Sch/U)_{fppf} \to \mathcal{X}$. Then $\{f(x) \to y\}$ is the desired fppf covering of $\mathcal{Y}$.

Proof of (2). Suppose that $y$ lies over the scheme $V$. We may think of $y$ as a morphism $(Sch/V)_{fppf} \to \mathcal{Y}$. By definition the 2-fibre product $\mathcal{X} \times_{\mathcal{Y}} (Sch/V)_{fppf}$ is representable by an algebraic space $W$ and the morphism $W \to V$ is surjective and smooth. Choose a scheme $U$ and a surjective étale morphism $U \to W$. Then $U \to V$ is also surjective and smooth (see Morphisms of Spaces, Lemmas 04XX, 03MG, and 03ZD). Hence $\{U \to V\}$ is a smooth covering. By More on Morphisms, Lemma 055V there exists an étale covering $\{V_i \to V\}$ such that each $V_i \to V$ factors through $U$. Denote $x_i$ the object of $\mathcal{X}$ over $V_i$ corresponding to the 1-morphism

$$(Sch/V_i)_{fppf} \to (Sch/U)_{fppf} \to \mathcal{X}.$$

Then $\{f(x_i) \to y\}$ is the desired étale covering of $\mathcal{Y}$. □